Exploring Key Factors Contributing to Students' Academic Success, Measured by Final Exam Score Using Machine Learning Models

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***Abstract*—With more emphasis on students' academic success, this research applies machine learning models like linear regression, ridge regression, random forest and XGBoost on personal, family and school factors of 6590 students aiming to find important predictors of exam scores and provide guidance to facilitate student's academic outcome. Linear regression model yields the highest R squared at the value of 0.77, suggesting a linear relationship between the independent variables and the target. The results emphasized that while higher exam scores are mostly determined by the students themselves, the effect of family and school should not be ignored. More personalized assistances should be provided to students to address potential obstacles and boost performance.**

Keywords—academic achievement, student motivation, family factors, school quality, parent involvement

I. Introduction

The demand for talents for modern development has spurred individuals, families and society at large to provide greater support for students with the aim of unlocking their full potential and fostering academic success. Previous studies have revealed that factors like students' motivation, attention and influence from family, school's role in providing personalised courses all contributes to the academic performance of students [1-3]. The current research seeks to find a suitable machine learning model to predict the exam scores of students which might capture potentially more complex relationship in data and interpret the results to provide insights on important factors affecting the performances of students. Relevant individuals and organisations may utilise this research to provide more targeted support programmes for students in need.

II. Data and Method

A. Data Source and Preparation

The Data is downloaded from Kaggle "Student Success: Factors & Insights" [4]. The original data has 6590 entries each representing a student. A total of 20 columns covering gender, study hours, school type and education level of parents are available.

The data provides a relatively rich range of variables which can potentially affect the exam score of students. In-school variables include *Attendence*, *Teacher_Quality*, *Extracurricular_Activities* (participation recorded as true and false) etc. Out-school variables includes *Sleeping_Hours*, *Family_Income* and *Distance_to_School* etc. The data takes family background and students' lifestyle and social life (measured by *Peer_Influence*) into consideration acknowledging the school alone is not the only decisive cause of students' scores and the relevant factors can be more complex from the student, the family and the school together.

Duplications and missing values were checked first. Missing values were found in three categorical variables: *Teacher_Quality*, *Parental_Education_Level* and *Distance_from_Home*. The missing values were imputed by the most common values in each variables.

By applying IQR methods and box plots, outliers were detected in *Hours_Studied* (43), *Tutoring_Sessions* (430) and *Exam_Score* (104). One entry in *Exam_Score* was found with a value 101 which could be result of error based on the general standard of full score being 100. This value was manually changed to 100. However, after inspecting the data flagged as outliers, no unrealistic value was found. For instance, students with tutorial session for more than 3 times but no more than 8 times were labelled as outliers yet such deviation should still be considered acceptable despite most students had fewer than 4 sessions. Thus, no "outlier" was removed.

B. Visualization

More information on numerical variables was revealed by creating histograms.

Most students studied about 15-25 hours per week and it was rare to study for less than 10 hours(with the minimum values at 1) or more than 30 hours(withe the maximum value at 44). A relatively normal distribution is witnessed for this variable *Hours_Studied*. All students managed to reach over 60% attendance rate. Most students sleep for at least 7 hours a day. Most students has no more than 2 tutoring session possibly due to high expenses. Most students has 2-4 hours of daily physical activities.

As we compare the distribution of previous scores to exam scores, it is found that the number of students with relatively good score (>= 80) and relatively low score(<= 60) significantly dropped in the current exam(target). In the previous exam, there were more than 300 students who got (almost) the full score. But in the current exam, most students scored around 70 and almost no student scored over 80. The difference between the exam scores potentially due to increased exam difficulty could potentially harm the model as it is intuitively expected that most students should have relatively stable performance and scores from new exams should not deviate much from previous exams. This issue will be elaborated in Discussion section later.

Based on the correlation heatmap, it makes sense that students who invest more time, attend courses more actively, score high previously and receiving more help from tutors can expect higher scores.

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For categorical variables, there were more male students than female student but the difference is not extreme. In general, having medium level of parental involvement, access to resources, motivation level and teacher quality were common. More students attended public schools, lived close to schools and had low or medium family income. The most common education level of parents were high school. Most student did not report having negative influence from peers. About 8% of students did not have access to Internet. Only about 10% of students reported having study disabilities. Even for students with high family income, there is no pattern of them enrolled in private schools more. Regardless of family income level, 70% of students still attended public schools. The teacher quality distribution has the same pattern in both public and private schools.

C. Model Selection and Evaluation

Linear regression was used as the baseline model due to its simplicity. Linear regression assumes linearity and additivity. Ridge regression introduces regularization to suffer from some increase in bias for a reduction in variance.

Ridge penalizes the squared value of coefficients by shrinking the coefficients but never to 0 (no feature gets removed). Therefore, it is useful when many features are expected to have very small effects and multicollinearity exists.

Random Forest and XGBoost are both ensemble methods using multiple tree methods. Random Forest builds many trees independently and takes the average prediction as the final prediction for the regression task while XGBoost builds many trees sequentially and uses the new tree to improve (correct the errors) the previous tree [5,6]. Both methods are suitable for complex relationships in the data.

Standard scaling was used on numerical values for linear regression and ridge regression. While scaling is not mandatory for linear regression, this step improves interpretation of coefficients. For ridge regression, without scaling, factors with larger numerical scale could bias the model by receiving more penalty.

GridSearch are used to tune the ridge regression and Random Forest to reach their expected best performance. Considering the current data size and model complexity, the greedy approach of trying out all combines given a few set values in selected parameters is acceptable. However, for XGBoost, the number of parameters and potential values creates more pressure on computation so RandomizedSearch was used. It takes only a limited number(specified using n_iter) of potential samples to test. While it can be argued that the best parameter here might not be the actual best but this tuning test can take forever if we really want a "best" result and it is worth it.

Considering the current regression task, Mean Squared Error (MSE), R squared and residual plot are used for evaluation.

III. Results

Linear regression model is the best model considering a Mean Squared Error at 3.26 and R squared at 0.77. Ridge regression (after tuning, with a small penalty) performed similarly as linear regression. This is because there is no strong correlation between variables. Ridge regression can improve the performance when multicollinearity exist and when we have many features but we expect most of them has little (not 0) contribution to the target.

Random Forest performed worse than linear regression with MSE at 4.38. The relationship in the data was likely mostly linear but Random Forest tried too hard to learn the patterns in the training data (expecting the relationship to be more complicated) and ended up with overfitting issue. XGBoost performed slightly worse than linear regression with MSE at 3.40 and R squared at 0.76 but still better than Random Forest due to the introduction of regularization to avoid overfitting issue.

Based on the coefficient of linear regression, for better academic outcome, the student will need high attendance, more time invested in studying, higher previous score, more tutoring session, high parental involvement, high access to resources, participation in extracurricular activities, living closer to school (potentially reduce commuting time and increase time for study and rest), positive peer influence, high quality teacher, relatively high education level for parents (at least college degree), high family income (medium is not enough), high motivation and no learning disabilities. By applying SHAP (summary plot and dependence plot) for interpreting XGBoost, we can find reach similar conclusion but the simplest model can already perform decently so there is no need to apply more complicated models.

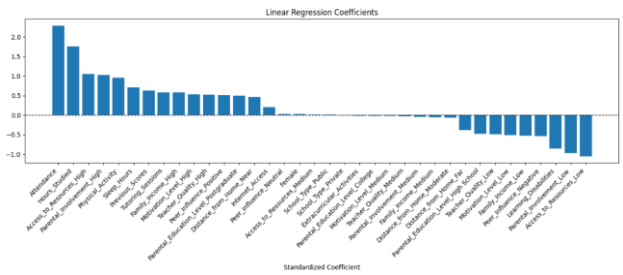

Fig. 1. Linear Regression Coefficients Visualization

IV. Discussion

The relationship between the current independent variables and the target is in general linear so linear regression is the best model.

Further research can be conducted with more variables and more samples. As was mentioned in previous section, a great difference between the distribution of the previous exam scores and that of current exam score (the target) was witnessed. More previous exam scores could be combined to create a new variable of average score of previous exams to show a more stable measure of previous academic performances.

While the model predicts exam scores as a measure of academic success, the usage of the model should be that based on the predicted score and the given variable values of an individual, more communication and collaboration can be promoted at the individual, family, school and society levels.

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